some deeper fundamental theory, it might drastically change our outlook. Maybe, as Johnson suggests, connections between large and small scales could imply a new fate for the cosmos. Or maybe, if we were able to revise quantum mechanics, we would finally find an explanation for dark energy. Even if we settle on a cosmological constant and a Heat Death future, according to Arkani-Hamed, we will still need a major shift on the theory side to be able to talk about what quantum fluctuations might do then, in terms of Boltzmann Brains or Poincaré recurrences. “In my mind it’s very unlikely that all these things are explained and understood within the framework of quantum mechanics,” he says. “I think we need some extension of quantum mechanics to help us talk about it.”

To what extent there even exists an explanation for the nature of our universe is an open question too. In the last decade or so, physicists have been grappling with the concept of the *landscape*—a theoretical multiverse of different possible spaces which could have drastically different conditions from our own. If such a landscape really exists, it could mean that the properties of the space we live in are merely environmental, rather than being set by some deep principle we haven’t been clever enough yet to find. This kind of multiverse can arise out of certain versions of inflation where new bubble universes inflate out of some eternal pre-existing space forever. “The idea that we’re the unique solution of the world does not seem right to me,” says Arkani-Hamed. “But on the other hand, when you try and make sense of the landscape and eternal inflation and all that stuff, it’s such a morass that I think that the whole conception of the problem is wrong to begin with.” Even with a landscape of possible universes, the basic problem still remains. “These questions about how to apply quantum mechanics to cosmology have been there almost from day zero. They’re not new. They were very difficult fifty years ago; they’re very difficult now.”

“I very firmly believe what we should be doing is actually just retracing our steps,” says Neil Turok, a cosmology theorist who has been looking at alternatives to cosmic inflation and who spent many years as the director of the Perimeter Institute for Theoretical Physics in Canada. “Go back, rewind fifty years and say, ‘Guys, we’re building on sand.’”

THE LONG VIEW